Examiner only

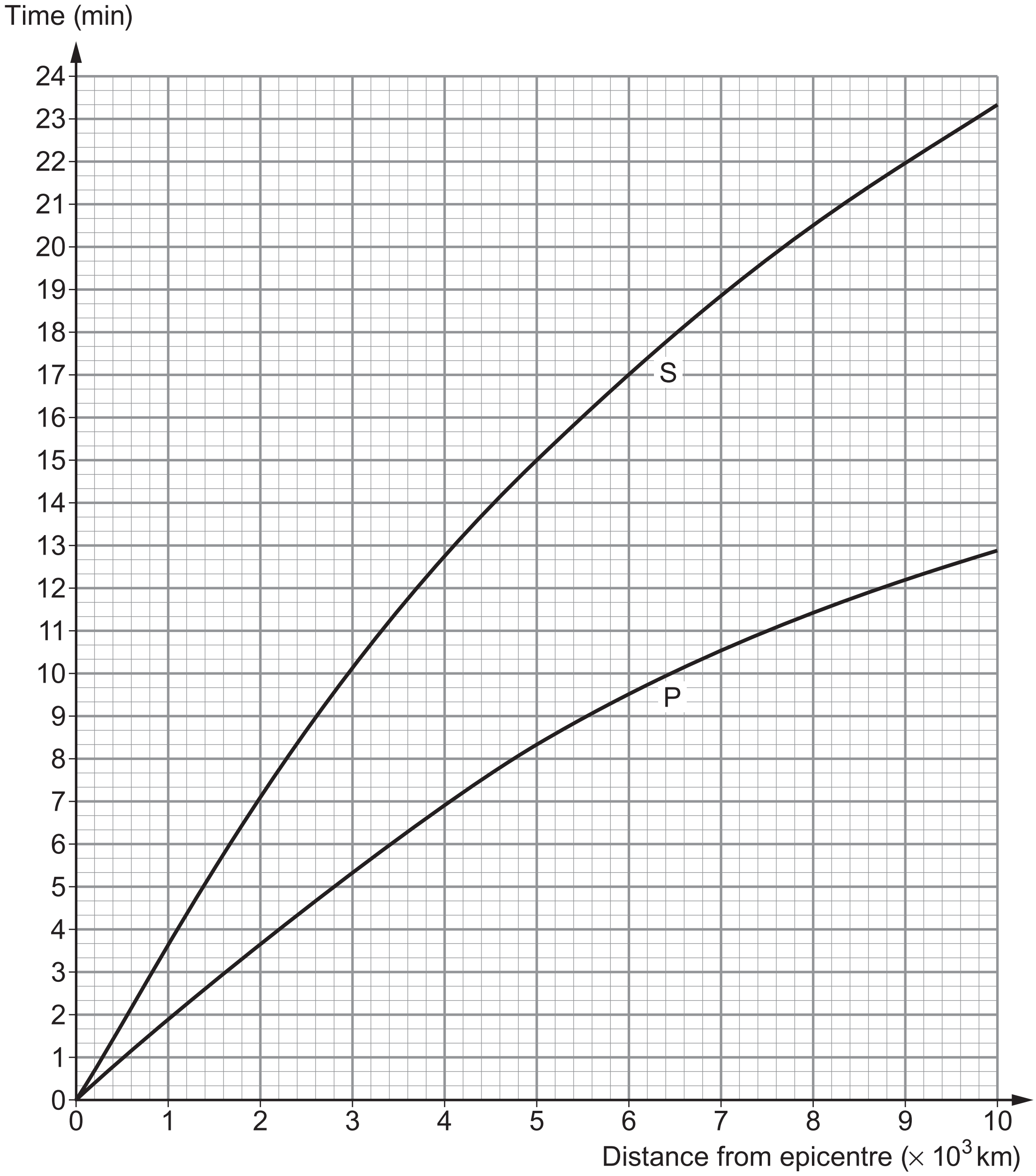
8. The ***epicentre*** is the point on the Earth's surface directly above an earthquake. Seismic stations detect earthquakes by the tracings made on seismographs.

(a) Surface, P and S waves are three types of earthquake waves.
Tick (✓) the boxes next to the **three** correct statements about earthquake waves. [3]

Surface waves travel the fastest	☐
S waves travel on the surface of the Earth	☐
S waves are transverse waves	☐
P waves travel through solids and liquids	☐
P waves are longitudinal waves	☐
S waves cause the most damage	☐

Examiner only

(b) The graph shows the time taken by P and S waves to travel different distances from the epicentre.



Each small square on the time axis represents 20 s.

(i) Use the graph to answer the following questions.

I. State the time it takes for a P wave to travel 5×10^3 km from the epicentre. [1]

time = ... min .... s

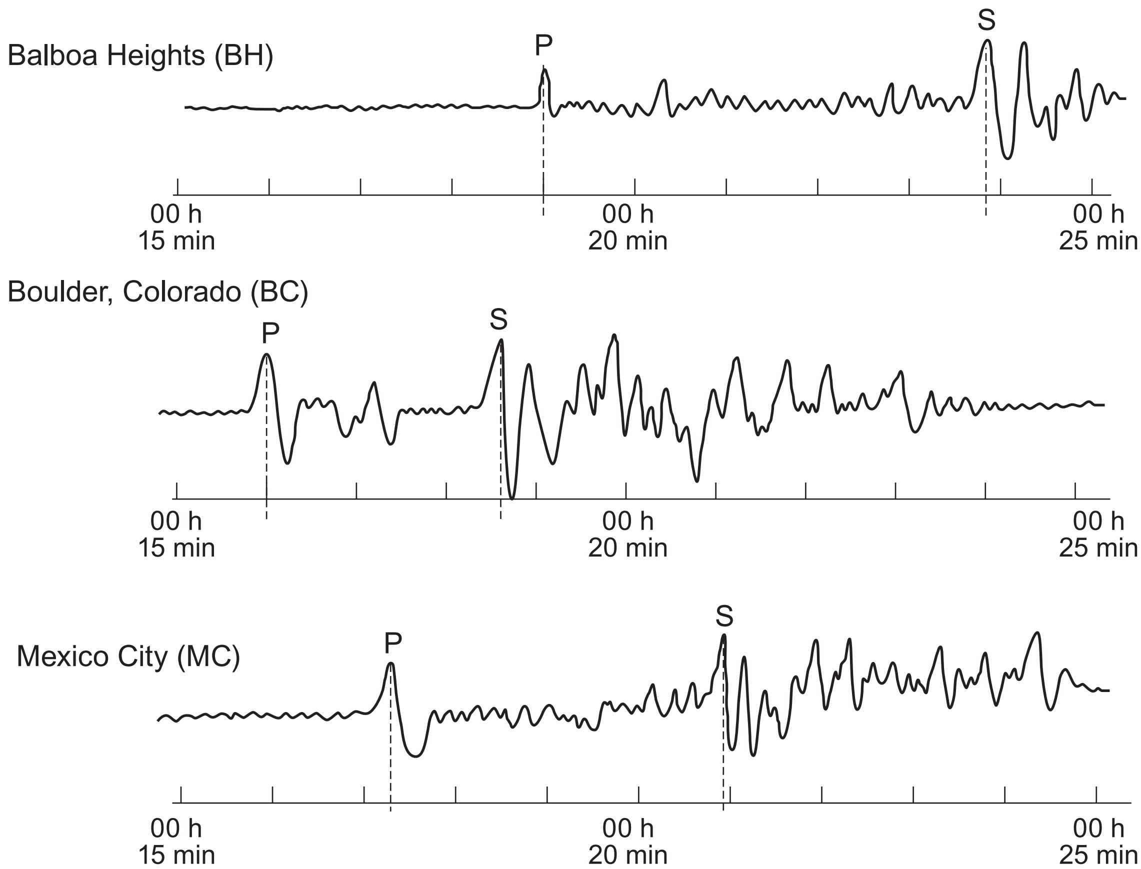
II. State the **extra** time it takes S waves to travel 5×10^3 km from the epicentre. [1]

time = ... min .... s

Sep Year 10 FT 2018_Q8

Examiner only

(ii) Study the three seismograph tracings below. Tracings made at three separate seismic stations are needed to locate an earthquake epicentre. P shows the arrival of P waves and S shows the arrival of S waves.



Use the information in the graph and tracings to **complete the table**. [3]

City	Arrival time of P waves (h:min:s)	Arrival time of S waves (h:min:s)	Time difference for P and S waves (h:min:s)	Distance to epicentre ($\times 10^3$ km)
Balboa Heights (BH)	00:19:00	00:23:50	00:04:50	3.2
Boulder, Colorado (BC)	.... : .... :	00:18:40	.... : .... :	....
Mexico City (MC)	00:17:15	00:20:55	00:03:40	2.2

TURN OVER FOR THE REST OF THE QUESTION

Examiner only

(iii) The P waves arriving at Balboa Heights (BH) took 6 **minutes** to travel from the epicentre. Use an equation from page 2 to calculate the speed of the P waves arriving at Balboa Heights in **km/s**. [Note that 3.2×10^3 km = 3 200 km] [2]

Speed = ... km/s

(iv) The data is used to locate the epicentre of the earthquake. **Indicate with crosses (X) on the diagram opposite two** possible positions for the location of the earthquake. [1]

(v) Use the data for Boulder Colorado (BC) to show clearly **on the diagram opposite** the actual location of the epicentre. Justify how you have arrived at your answer. [2]

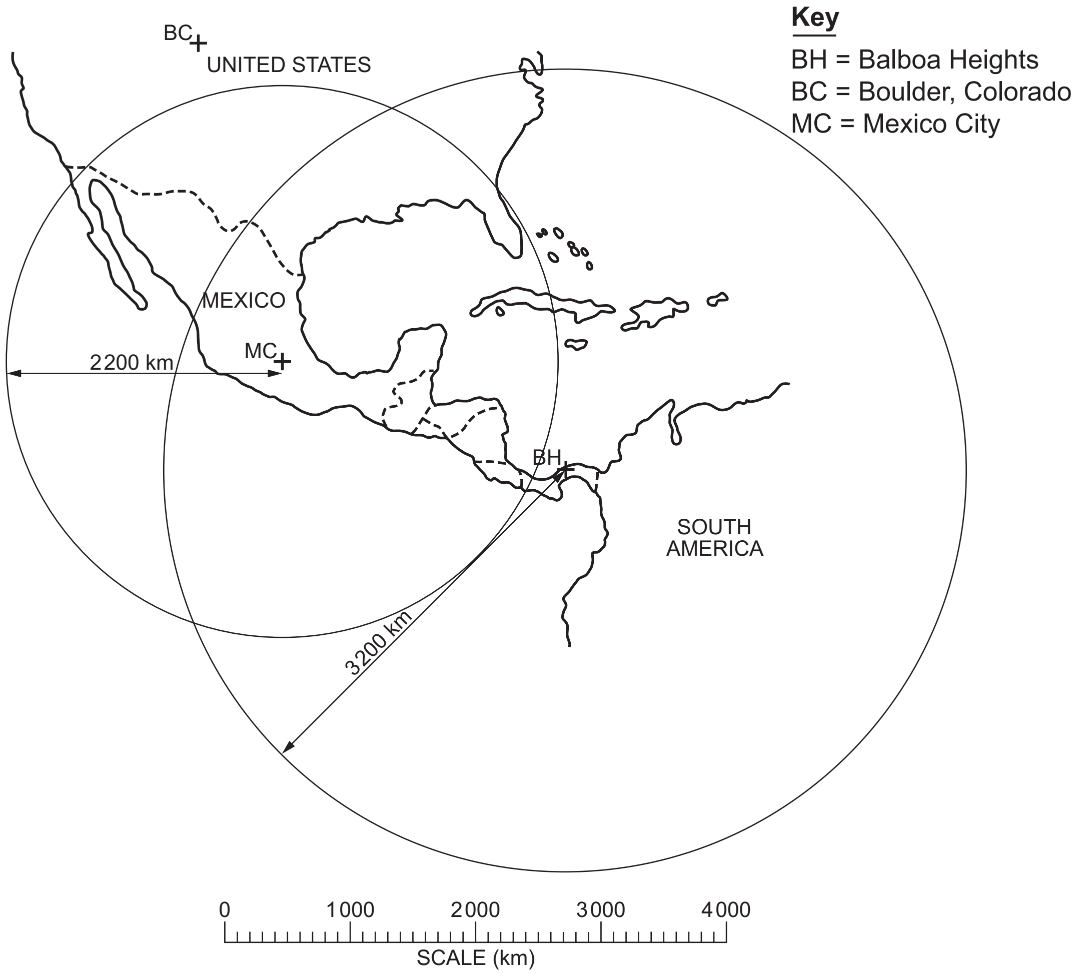
....

....

....

Sep Year 10 FT 2018_Q8

Examiner only



13

END OF PAPER

 (3420U10-1) **Turn over.**